

One Factor Models

Linear models with only categorical predictors (or factors) have traditionally been called analysis of variance (ANOVA) problems. The idea is to partition the overall variance in the response due to each of the factors and the error. This traditional approach is exemplified by Scheffé (1959).

Although some insight can be gained by considering the problem from this perspective, the price is an increasingly complex set of specialized formulae required for each type of model depending on the configuration of factors. Instead, we take a regression-based approach by putting the model into the $y = X\beta + \varepsilon$ format and then using the inferential methods we have already developed.

The terminology used in ANOVA-type problems is sometimes different. Predictors are now all qualitative and are now typically called *factors*, which have some number of *levels*. The regression parameters are now often called *effects*. We shall consider only models where the parameters are considered fixed, but unknown — called *fixed-effects* models. *Random-effects* models are used where parameters are taken to be random variables and are not covered in this text.

15.1 The Model

Suppose we have a factor α occurring at $i = 1, \dots, I$ levels, with $j = 1, \dots, J_i$ observations per level. We use the model:

$$y_{ij} = \mu + \alpha_i + \varepsilon_{ij}$$

The parameters are not identifiable. For example, we could add some constant to μ and subtract the same constant from each α_i and the fit would be unchanged. Some restriction is necessary. Here are some possibilities:

1. Drop μ from the model and use I different dummy variables to estimate α_i for $i = 1, \dots, I$. This is feasible but does not extend well to models with more than one factor as more than one parameter needs to be dropped.
2. Set $\alpha_1 = 0$, then μ represents the expected mean response for the first level and α_i for $i \neq 1$ represents the difference between level i and level one. Level one is then called the *reference level* or *baseline level*. This can be achieved using treatment contrasts as discussed in the previous chapter.
3. Set $\sum_i \alpha_i = 0$, now μ represents the mean response over all levels and α_i , the difference from that mean. This requires the use of sum contrasts.

The choice of constraint from those listed above or otherwise will determine the coding used to generate the X -matrix. Once that is done, the parameters (effects) can be estimated in the usual way along with standard errors. No matter which valid constraint and coding choice is made, the fitted values and residuals will be the same.

Once the effects are estimated, the natural first step is to test for differences in the levels of the factor. An explicit statement of the null and alternative hypotheses would depend on the coding used. If we use the treatment coding with a reference level, then the null hypothesis would require that $\alpha_2 = \dots = \alpha_I = 0$. For other codings, the statement would differ. It is simpler to state the hypotheses in terms of models:

$$\begin{aligned} H_0: y_{ij} &= \mu + \varepsilon_{ij} \\ H_1: y_{ij} &= \mu + \alpha_i + \varepsilon_{ij} \end{aligned}$$

We compute the residual sum of squares and degrees of freedom for the two models and then use the same F-test as we have used for regression. The outcome of this test will be the same no matter what coding/restriction we use. If we do not reject the null, we are almost done — we must still check for a possible transformation of the response and outliers. If we reject the null, we must investigate which levels differ.

15.2 An Example

Twenty-four animals were randomly assigned to four different diets and blood samples were taken in a random order. The blood coagulation time was measured. These data come from Box et al. (1978):

```
> data(coagulation, package="faraway")
> coagulation
  coag diet
1    62   A
2    60   A
...etc...
23   63   D
24   59   D
```

Some preliminary graphical analysis is essential before fitting. The default plotting method in base R graphics is the side-by-side boxplot. Although this plot has strong visual impact for comparison, a stripchart can be better for smaller datasets:

```
> plot(coag ~ diet, coagulation, ylab="coagulation time")
> stripchart(coag ~ diet, coagulation, vertical=TRUE, method="stack",
  xlab="diet", ylab="coagulation time")
```

The boxplot in Figure 15.1 shows how the four levels vary but there is something odd about the display of diet C because the median and upper quartile are the same. The stripchart shows the ties in the data in diets C and D.

We are looking for several features in these plots. We must check for equality of variance in the groups, which seems satisfied in this example. We are looking for evidence of skewness showing a lack of normality. This might suggest a transformation of the response. There is no such concern in this example. Finally, we should look for outliers — there are none to be seen in this example.

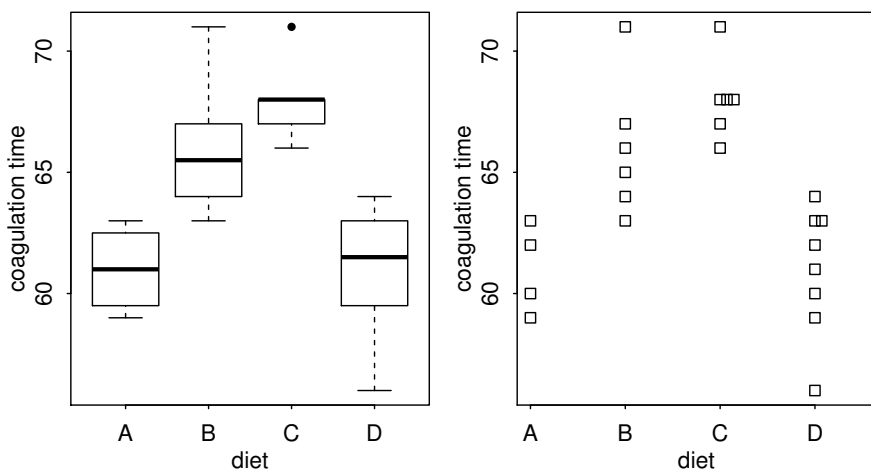


Figure 15.1 A boxplot on the left and a stripchart on the right showing the blood coagulation data.

It is not worth considering diagnostics for influence, as the leverages depend explicitly on J_i . If there is only one observation for a given level, that is $J_i = 1$, then the estimate of the effect for that level will be based on that single point. That point is clearly influential without further investigation.

Now let's fit the model using the default treatment coding:

```
> lmod <- lm(coag ~ diet, coagulation)
> sumary(lmod)
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	6.10e+01	1.18e+00	51.55	< 2e-16
dietB	5.00e+00	1.53e+00	3.27	0.00380
dietC	7.00e+00	1.53e+00	4.58	0.00018
dietD	2.99e-15	1.45e+00	0.00	1.00000

n = 24, p = 4, Residual SE = 2.366, R-Squared = 0.67

Rounding error results in a rather unfortunate formatting of the results. The coefficients can be more cleanly seen as:

```
> round(coef(lmod), 1)
```

	dietB	dietC	dietD
(Intercept)	61	5	7

Group A is the reference level and has a mean of 61, groups B, C and D are 5, 7 and 0 seconds larger, respectively, on average. Examine the design matrix to understand the coding:

```
> model.matrix(lmod)
```

The three test statistics for the group levels correspond to comparisons with the reference level A. Although these are interesting, they do not specifically answer the

question of whether there is a significant difference between any of the levels. This test can be obtained as:

```
> anova(lmod)
Analysis of Variance Table

Response: coag
          Df Sum Sq Mean Sq F value    Pr(>F)
diet        3    228    76.0    13.6 0.000047
Residuals  20    112     5.6
```

We see that there is indeed a difference in the levels although this test does not tell us which levels are different from others.

We can fit the model without an intercept term as in:

```
> lmodi <- lm(coag ~ diet -1, coagulation)
> summary(lmodi)
      Estimate Std. Error t value Pr(>|t|)
dietA      61.000      1.183   51.5    <2e-16
dietB      66.000      0.966   68.3    <2e-16
dietC      68.000      0.966   70.4    <2e-16
dietD      61.000      0.837   72.9    <2e-16

n = 24, p = 4, Residual SE = 2.366, R-Squared = 1
```

We can directly read the level means. The R^2 is not correctly calculated because of the absence of an intercept. To generate the usual test that the means of the levels are equal, we would need to fit the null model and compare using an F-test:

```
> lmnul1 <- lm(coag ~ 1, coagulation)
> anova(lmnul1, lmodi)
Analysis of Variance Table

Model 1: coag ~ 1
Model 2: coag ~ diet - 1
  Res.Df RSS Df Sum of Sq    F    Pr(>F)
1      23 340
2      20 112  3      228 13.6 0.000047
```

We get the same F-statistic and p-value as in the first coding.

We can also use a sum coding:

```
> options(contrasts=c("contr.sum", "contr.poly"))
> lm1 <- lm(coag ~ diet, coagulation)
> summary(lm1)
      Estimate Std. Error t value Pr(>|t|)
(Intercept)   64.000      0.498  128.54    < 2e-16
diet1         -3.000      0.974   -3.08    0.00589
diet2          2.000      0.845    2.37    0.02819
diet3          4.000      0.845    4.73    0.00013

n = 24, p = 4, Residual SE = 2.366, R-Squared = 0.67
```

So the estimated overall mean response is 64 while the estimated mean response for A is three less than the overall mean, that is, 61. Similarly, the means for B and C are 66 and 68, respectively. Since we are using the sum constraint, we compute

$\hat{\alpha}_D = -(-3 + 2 + 4) = -3$ so the mean for D is $64 - 3 = 61$. Notice that $\hat{\sigma}$ and the R^2 are the same as before.

So we can use any of these three methods and obtain essentially the same results. Dropping the intercept is least convenient since an extra step is needed to generate the F-test. Furthermore, the approach would not extend well to experiments with more than one factor, as additional constraints would be needed. The other two methods can be used according to taste. The treatment coding is most appropriate when the reference level is set to a possible control group. We will use the treatment coding by default, as does R, for the rest of this book.

15.3 Diagnostics

There are fewer diagnostics to do for ANOVA models, but it is still important to plot the residuals and fitted values and to make the Q-Q plot of the residuals. It makes no sense to transform the predictor, but it is reasonable to consider transforming the response. Diagnostics are shown in Figure 15.2.

```
> qqnorm(residuals(lmod))
> qqline(residuals(lmod))
> plot(jitter(fitted(lmod)), residuals(lmod), xlab="Fitted", ylab="
  Residuals")
> abline(h=0)
```

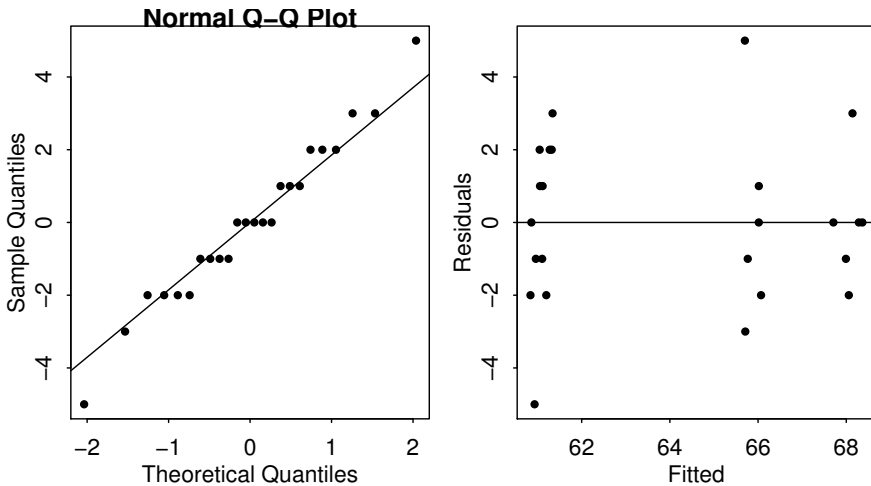


Figure 15.2 Diagnostics for the blood coagulation model.

Because the data are integers and the fitted values happen to be integers also, some discreteness is obvious in the Q-Q plot. Of course, discrete data cannot be normally distributed. However, here the residuals are approximately normal and so we can go ahead with the inference without much concern. The discreteness in the residuals and fitted values shows up in the residual-fitted plot. We have jittered the points so that they can be seen separately.

The assumption of homogeneity of the error variance can be examined using Levene's test. It computes the absolute values of the residuals and uses these as the response in a new one-way ANOVA. To reduce the possible influence of outliers, group medians rather than means should be used. A significant difference would indicate non-constant variance.

Most tests and confidence intervals (CIs) are relatively insensitive to non-constant variance so there is no need to take action unless the Levene test is significant at the 1% level.

Applying this to the diet data, we find:

```
> med <- with(coagulation, tapply(coag, diet, median))
> ar <- with(coagulation, abs(coag - med[diet]))
> anova(lm(ar ~ diet, coagulation))
```

Analysis of Variance Table

Response: ar

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
diet	3	4.3	1.4	0.65	0.59
Residuals	20	44.5	2.2		

Since the p-value is large, we conclude that there is no evidence of a non-constant variance.

An alternative test is due to Bartlett:

```
> bartlett.test(coag ~ diet, coagulation)
```

Bartlett test of homogeneity of variances

data: coag by diet

Bartlett's K-squared = 1.668, df = 3, p-value = 0.6441

Again, no difference is found. Levene's test is more robust to outliers.

15.4 Pairwise Comparisons

After detecting some difference in the levels of the factor, interest centers on which levels or combinations of levels are different. It does not make sense to ask whether a particular level is significant since this begs the question of "significantly different from what?" Any meaningful test must involve a comparison of some kind.

A pairwise comparison of level i and j can be made using a CI for $\alpha_i - \alpha_j$ using:

$$\hat{\alpha}_i - \hat{\alpha}_j \pm t_{df}^{\alpha/2} se(\hat{\alpha}_i - \hat{\alpha}_j)$$

where $se(\hat{\alpha}_i - \hat{\alpha}_j) = \hat{\sigma} \sqrt{1/J_i + 1/J_j}$ and $df = n - I$ in this case. A test for $\alpha_i = \alpha_j$ amounts to seeing whether zero lies in this interval or not. For example, let's find a 95% CI for $B - A$. From the model output, we can see that the difference is 5.0 with a standard error of 1.53. For differences not involving the reference level of A, more effort would be required to calculate these values. The interval is:

```
> 5 + c(-1, 1) * qt(0.975, 24-4) * 1.53
[1] 1.8085 8.1915
```

Since zero is not in the interval, the difference is significant. This is fine for just one test, but we are likely to be interested in more than one comparison. Suppose we do all possible pairwise tests when $\alpha = 5\%$ and the null hypothesis is in fact true. In the blood coagulation data, there are four levels and so six possible pairwise comparisons. Even if there was no difference between the four levels, there is still about a 20% chance that at least one significant difference will be found.

For experiments with more levels, the true type I error gets even higher. Using the t-based CIs for multiple comparisons is called the least significant difference (LSD) method, but it can hardly be recommended. Now one might be tempted to argue that we could choose which comparisons are interesting and so reduce the amount of testing and thus the magnitude of the problem. If we only did a few tests, then the Bonferroni adjustment (see Section 6.2.2) could be used to make a simple correction. However, the determination of which comparisons are “interesting” is usually made after seeing the fitted model. This means that all other comparisons are implicitly made even if they are not explicitly computed. On the other hand, if it can be argued that the comparisons were decided before seeing the fit, then we could make the case for the simple adjustment. However, this is rarely the case and furthermore it might be difficult to convince others that this really was your intention. We must usually find a way to adjust for *all* pairwise comparisons.

There are many ways to make the adjustment, but *Tukey’s honest significant difference (HSD)* is the easiest to understand. It depends on the studentized range distribution which arises as follows. Let $X_1, \dots, X_n$ be i.i.d. $N(\mu, \sigma^2)$ and let $R = \max_i X_i - \min_i X_i$ be the range. Then $R/\hat{\sigma}$ has the studentized range distribution $q_{n,v}$ where v is the number of degrees of freedom used in estimating σ .

The Tukey CIs are:

$$\hat{\alpha}_i - \hat{\alpha}_j \pm \frac{q_{l,df}}{\sqrt{2}} \hat{\sigma}(1/J_i + 1/J_j)$$

When the level sample sizes J_i are very unequal, Tukey’s HSD test may become too conservative. We compute the Tukey HSD bands for the B-A difference. The critical value from the studentized range distribution is obtained from the `qtukey()` function:

```
> se <- summary(lmod)$sigma
> 5 + c(-1,1) * qtukey(0.95, 4, 24-4)/sqrt(2) * se * sqrt(1/4 + 1/6)
[1] 0.72455 9.27545
```

A convenient way to obtain all the intervals is:

```
> (tci <- TukeyHSD(aov(coag ~ diet, coagulation)))
Tukey multiple comparisons of means
 95% family-wise confidence level

Fit: aov(formula = coag ~ diet, data = coagulation)

$diet
      diff      lwr      upr    p adj
B-A      5  0.72455  9.2754 0.01833
C-A      7  2.72455 11.2754 0.00096
```

D-A	0	-4.05604	4.0560	1.00000
C-B	2	-1.82407	5.8241	0.47660
D-B	-5	-8.57709	-1.4229	0.00441
D-C	-7	-10.57709	-3.4229	0.00013

We find that only the $A - D$ and $B - C$ differences are not significant as the corresponding intervals contain zero. The duality between confidence intervals and hypothesis testing can also be used to produce an adjusted p -value as seen in the final column of the output. This merely confirms our impression of which differences are significant. The intervals can be plotted as seen in Figure 15.3.

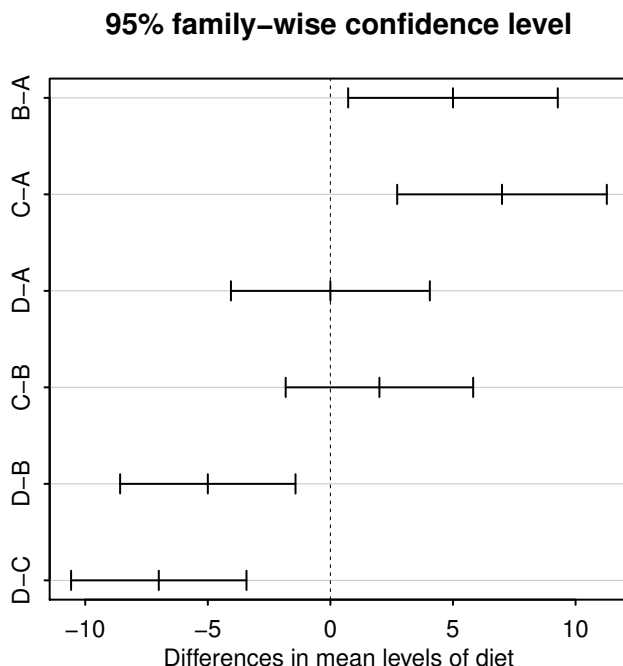


Figure 15.3 Tukey HSD 95% confidence intervals for the pairwise differences.

```
> plot(tci)
```

The Tukey method assumes the worst by focusing on the largest difference. There are other competitors like the Newman-Keuls, Duncan's multiple range and the Waller-Duncan procedure, which are less pessimistic or do not consider all possible pairwise comparisons. For a detailed description of the many available alternatives see Bretz, Hothorn, and Westfall (2010) and the R package `multcomp`.

15.5 False Discovery Rate

Consider some data taken from the Junior School Project collected from primary (US term is elementary) schools in inner London. The data is described in detail in

Mortimore et al. (1988). We focus on just two of the variables in the data — the `school`, of which there are 49, and the mathematics test scores for students from these schools. Suppose we are interested in deviations from the average and so we center the scores:

```
> data(jsp, package="faraway")
> jsp$mathcent <- jsp$math - mean(jsp$math)
```

A more pleasing plot of the data can be obtained using the `ggplot2` package as seen in Figure 15.4. We have to rotate the school labels so they can be distinguished.

```
> require(ggplot2)
> ggplot(aes(x=school,y=mathcent),data=jsp)+geom_boxplot() + theme(
  axis.text.x = element_text(angle = 90))+ylab("Centered Math
  Scores")
```

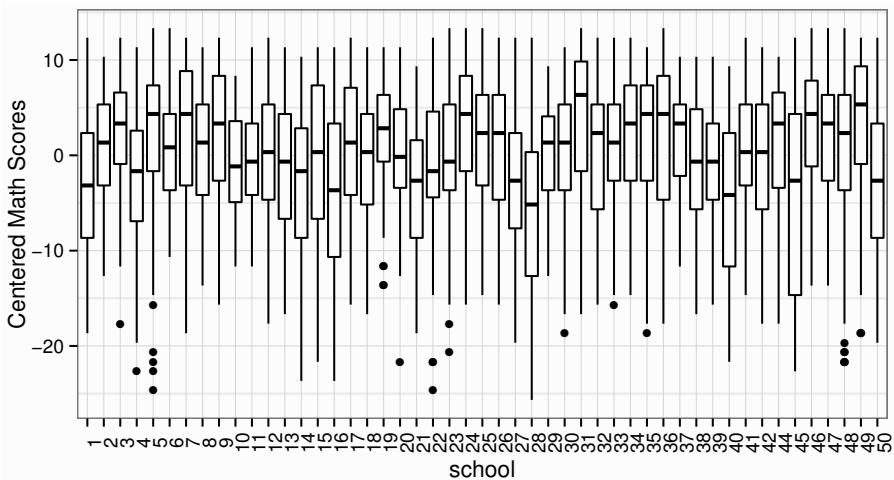


Figure 15.4 Variation-centered math scores by school.

Let's choose the parameterization that omits the intercept term:

```
> lmod <- lm(mathcent ~ school-1, jsp)
> sumary(lmod)
```

	Estimate	Std. Error	t value	Pr(> t)
school1	-3.3685	0.7686	-4.38	1.2e-05
school2	0.6714	1.2287	0.55	0.58481
...				
school150	-2.6520	0.7336	-3.62	0.00030

```
n = 3236, p = 49, Residual SE = 7.372, R-Squared = 0.08
```

Since we have centered the response, the t-tests, which check for differences from zero, are meaningful. We can see there is good evidence that schools 1 and 50 are significantly below average while the evidence that school 2 is above average is not statistically significant. We can test for a difference between the schools:

```
> anova(lm(mathcent ~ school, jsp))
Analysis of Variance Table

Response: mathcent
      Df Sum Sq Mean Sq F value Pr(>F)
school  48  15484      323    5.94 <2e-16
Residuals 3187 173212      54
```

We find a strongly significant difference. This comes as little surprise as the sample size is large, giving us the power to detect quite small differences. Furthermore, we may have strong prior reasons to expect differences between the schools. A more interesting question is which schools show clear evidence of under- or over-performance?

There are too many pairwise comparisons on which to focus our interest. Instead let us ask which schools have means significantly different from the average. The parameterization we have chosen makes these comparisons easy but we would expect about 5% of these differences to be significant even if the null hypothesis held.

Some adjustment is necessary. One approach is to control the *familywise error rate* (FWER) which is the overall probability of falsely declaring a difference (where none exists). The *Bonferroni correction* is a simple way to do this. We just multiply the unadjusted p -values by the number of comparisons. Any probability computed above one is truncated to one. Let's see which schools have adjusted p -values less than 5%:

```
> pvals <- summary(lmod)$coef[,4]
> padj <- p.adjust(pvals, method="bonferroni")
> coef(lmod)[padj < 0.05]
school11 school116 school121 school128 school131 school140 school145
-3.3685   -3.7374   -3.6185   -5.8286    3.8241   -4.9855   -4.6392
school150
-2.6520
```

We see that eight schools are identified with all except school 31 marked as significantly below average.

The Bonferroni correction is known to be conservative but even were we to use one of the more generous alternatives, the familywise error rate restriction imposes a high bar on the identification of significant effects. As the number of levels being compared increases, this requirement becomes ever more stringent.

An alternative approach is to control the *false discovery rate* (FDR) which is the proportion of effects identified as significant which are not real. The best known method of doing this is due to Benjamini and Hochberg (1995).

Given sorted p -values $p_{(i)}$ for $i = 1, \dots, m$ the procedure finds the largest i for which $p_{(i)} \leq \alpha i/m$. All tests corresponding to $p_{(i)}$ up to and including this i are declared significant. We compute this for our example:

```
> names(which(sort(pvals) < (1:49)*0.05/49))
[1] "school128" "school131" "school121" "school11" "school145" "school140"
[7] "school116" "school150" "school147" "school149" "school14" "school136"
[13] "school146" "school114" "school124" "school127" "school134" "school19"
```

We see that 18 schools are identified compared to the 8 by the previous procedure. FDR is less stringent than FWER in identifying significant effects. A more convenient method of computing the adjusted p -values is:

```
> padj <- p.adjust(pvals, method="fdr")
> coef(lmod)[padj < 0.05]
```

school11	school14	school19	school114	school116	school121	school124
-3.3685	-2.6619	2.2458	-3.2898	-3.7374	-3.6185	2.6238
school127	school128	school131	school134	school136	school140	school145
-2.3058	-5.8286	3.8241	1.9359	2.3605	-4.9855	-4.6392
school146	school147	school149	school150			
3.0636	2.3969	2.7964	-2.6520			

FDR methods are more commonly used where large numbers of comparisons are necessary as often found in imaging or bioinformatics applications. In examples such as these, we expect to find some significant effects and FDR is a useful tool in reliably identifying them.

Exercises

- Using the `pulp` data, determine whether there are any differences between the operators. What is the nature of these differences?
- Determine whether there are differences in the weights of chickens according to their feed in the `chickwts` data. Perform all necessary model diagnostics.
- Using the `PlantGrowth` data, determine whether there are any differences between the groups. What is the nature of these differences? Test for a difference between the average of the two treatments and the control.
- Using the `infmort` data, perform a one-way ANOVA with `income` as the response and `region` as the predictor. Which pairs of regions are different? Now check for a good transformation on the response and repeat the comparison.
- The `anaesthetic` data provides the time to restart breathing unassisted in recovering from general anaesthetic for four treatment groups.
 - Produce a boxplot depicting the data. Comment on any features of interest.
 - Make an appropriate stripchart of the data.
 - Produce versions of the previous two plots using the `ggplot2` package. Show how you can overlay these two plots.
 - Fit a one-factor model for the recovery times and test for a difference between the two groups.
 - Try the Box-Cox transformation method. Explain what went wrong.
 - Try a square root transformation on the response. Are the diagnostics satisfactory? Is there a significant difference among the treatment groups?
- Data on the butterfat content of milk from Canadian cows of five different breeds can be found in the `butterfat` dataset. Consider only mature cows.
 - Plot the data and interpret what you see.
 - Test for a difference between the breeds.

- (c) Check the diagnostics for your model.
 - (d) Produce a plot to check for differences in butterfat between breeds. Which pairwise differences are not statistically significant?
7. Five suppliers cut denim for a jeans manufacturer. The amount of waste relative to a target was collected weekly as seen in the `denim` dataset.
- (a) Plot the data to determine which supplier wastes the least. Which supplier is best in terms in minimizing maximum weekly waste?
 - (b) Is there a significant difference in wastage between the suppliers?
 - (c) Check the regression diagnostics commenting on any violations.
 - (d) Remove two outliers and repeat the test for a significant difference. Which supplier has the lowest predicted wastage under this model?
 - (e) Check for significant pairwise differences between suppliers. Which pairs are significantly different?
 - (f) Which supplier would you pick if there was no other relevant difference between them? What if the cost of the suppliers was in numerical order with the first the most expensive and five the cheapest?